Even Wins (13.20)

A. Engel — Problem-Solving Strategies · 1998

Even Wins. Initially there is a supply of $(2n + 1)$ chips. A and B take turns to remove any number of chips from 1 to k. At the end, one of the players winds up with an even number of chips, the other with an odd number. The winner is the one who possesses an even number of chips. Find the losing position for (a) $k = 3$, (b) $k = 4$, (c) even k, (d) odd k.

Consider also the case that *Odd Wins*. See [28].